

Liam Concierge

Package assistance for residents
An AI application with verified actions

Arzhan Moldabayev · nFactorial Final Project

The package-room problem

Residents need to know what has arrived and when they can receive it.

An operator needs a reliable recipient, location and custody record.

Liam Concierge connects those tasks with scheduled delivery. The prototype supports one daily window and direct handoff.

A resident's delivery request

The resident sees their own packages and available windows.

A delivery request waits for explicit confirmation. Reloading the page preserves that step.

The server checks availability again before booking.

[← Package dashboard](#)[Expected packages](#)

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Liam Concierge · AI assistant

A little help, whenever you need it.

I can check your package records, explain service rules and help you choose a delivery window. Physical service follows the available windows.

How can I help?

My packages

What am I expecting?

Available windows

Request a delivery

The buttons work without AI processing.

Your message

Ask in English or Russian

☐ I agree to send this message to OpenAI for this request. Do not include passwords or sensitive personal information. Raw messages and package details are omitted from external traces.

Ask Liam Concierge

Your concierge

Your delivery request is confirmed for the selected window.

UPS · LIAM-DEMO-1002

Delivered

Recorded location: Shelf A · 03

Amazon · LIAM-DEMO-1001

Scheduled

Recorded location: Shelf A · 02

Sep 25, 12:47 AM – Sep 25, 1:47 AM ET

► Request steps

Recent requests

Request a delivery · complete

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LangGraph orchestration

Classify the request, then branch to package tools, service rules or an access refusal.

Booking reads packages and windows, persists an interrupt in PostgreSQL, and resumes after confirmation.

The final MCP tool consumes a short-lived approval once. SQL locks enforce capacity.

Eight tools through an actual MCP server

Tool	Access / purpose
get_parcels	Resident's physical and expected records
get_available_windows	Available booking windows
request_delivery	Explicit approval; transactional booking
get_package_details	Own package; operator can inspect all
get_room_capacity	Operator; timestamped room inspection
get_aging_packages	Operator; age since physical intake
get_package_exceptions	Operator; unresolved intake issues
preview_delivery_round	Operator; workload and recorded weights

MCP exposes typed, discoverable tools to the graph. Web APIs still handle browser requests and Gmail OAuth.

PDF retrieval with page citations

An eight-page demonstration policy provides the knowledge base.

PDF text becomes page-based chunks and 256-dimensional embeddings in pgvector. Retrieval selects up to four relevant chunks.

The answer cites allowed source IDs and PDF pages. Unsupported questions produce an explicit absence-of-information response.

Label reading with a runtime Skill

A phone image provides the visible name, unit, tracking reference and printed weight.

The app loads skills/liam-parcel-exceptions/SKILL.md into the vision request. Missing values stay empty.

A verified local roster proposes a recipient. The operator reviews the draft before physical intake.

FICTIONAL TEST LABEL - NOT FOR SHIPPING

Amazon

Recipient: Avery Stone

Unit: DEMO-101

Tracking: TBA880000000000

Weight: 1 lb

Residents choose what to share

Tracking-only entry works without mailbox access.

A resident can paste one delivery notice, or separately connect their own Gmail account with read-only permission.

Expected shipment records remain separate from physical custody. Accounts cannot read one another's notices.

Evaluation on 40 frozen examples

Task	Result	Cases
Vision fields	119 / 120 correct	20 labels
Complete label match	19 / 20	20 labels
Review branch	20 / 20	20 labels
Policy citation checks	10 / 10	10 questions
Intent classification	10 / 10	10 requests

One vision result added a digit to a long tracking number. All fixtures are synthetic; these results do not measure accuracy in a real mailroom.

Model and retrieval experiments

Comparison	Quality	Median latency
GPT-4.1 mini · intent	12 / 12	760 ms
GPT-6 Luna · intent	11 / 12	909 ms
Page chunks · RAG	10 / 10	2,566 ms
Fixed chunks · RAG	10 / 10	2,375 ms

Selected GPT-4.1 mini: temperature 0, top_p 1, output limit 512. Page chunks preserve context with 8 chunks versus 24. RAG timing includes the evaluation judge.

Observable calls and verified boundaries

Langfuse records graph steps, MCP calls, Skill loading, model settings, token usage and timing.

Explicit redaction keeps raw messages, images and package details out of external traces.

51 automated tests and the production browser flow passed, including real AI vision and PDF retrieval.

Demonstration scope and next steps

The working core covers intake review, personal package queries, policy answers and confirmed delivery requests.

Next: real-device label tests, Gmail authorization and stronger operational evaluation.

Three daily rounds, outbound reminders, body-camera footage and courier GPS remain outside this prototype.